Linux Log Analysis Quick Reference

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Linux powers most servers, cloud infrastructure, and security tools. Knowing where logs live, what they contain, and how to parse them quickly is essential for detecting brute force attacks, privilege escalation, malware, and lateral movement.

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1. Linux Log Locations

Log File	Path	What It Contains
Authentication log	/var/log/auth.log	SSH logins, sudo commands, user additions/deletions, password changes
System log	/var/log/syslog	General system messages, daemon info, hardware events
Kernel log	/var/log/kern.log	Kernel events, driver issues, hardware errors
Boot log	/var/log/boot.log	System startup messages
Package manager log	/var/log/dpkg.log or /var/log/yum.log	Software installations and updates
Apache access log	/var/log/apache2/access.log	Web server requests
Apache error log	/var/log/apache2/error.log	Web server errors
Nginx access log	/var/log/nginx/access.log	Web server requests
Nginx error log	/var/log/nginx/error.log	Web server errors
MySQL/MariaDB log	/var/log/mysql/error.log	Database errors and queries
Cron log	/var/log/cron.log	Scheduled task executions
Mail log	/var/log/mail.log	Email server activity
User commands log	/var/log/user.log	User-level command information
Failed login attempts	/var/log/btmp	Binary log of failed logins (use lastb command)
Successful logins	/var/log/wtmp	Binary log of all logins (use last command)
Current logins	/var/run/utmp	Currently logged-in users (use who, w commands)
Systemd journal	journalctl command	Centralised logging for systemd systems

2. Key Log Files in Detail

/var/log/auth.log — Most Important for Security

This log records all authentication events. This is your primary source for detecting brute force attacks, unauthorised access, and privilege abuse.

Message	What It Means	Severity
Failed password for root from 192.168.1.100	SSH brute force attempt	High
Invalid user test from 203.0.113.5	Reconnaissance (user doesn't exist)	High
Accepted password for user from 10.0.0.1	Successful SSH login	Info (investigate if unusual)
session opened for user root	Root login or sudo session	Critical

session closed for user	Logout	Info
sudo: user : TTY=pts/0 ; COMMAND=/bin/bash	Sudo command execution	Medium
password changed for user	Password change	Medium
useradd: new user added	New user created	High
userdel: user deleted	User deleted	High
groupadd: group added	New group created	Medium
Authentication failure	Generic auth failure	Medium

/var/log/syslog — General System Information

This log captures system-wide events from various services.

Message	What It Means	Severity
kernel: [UFW BLOCK]	Firewall blocked connection	Medium
CRON[1234]: (root) CMD	Scheduled task executed	Info
systemd: Failed to start service	Service failed to start	Medium
dhclient: DHCPACK from 192.168.1.1	IP address assigned	Info
sshd: Server listening on 0.0.0.0 port 22	SSH service started	Info

/var/log/kern.log — Kernel Events

This log captures low-level system events that can indicate compromise.

Message	What It Means	Severity
Out of memory: Kill process	System ran out of memory	High
Segmentation fault	Program crashed (possible exploit)	Medium
CPU soft lockup	System frozen or overloaded	Medium
New USB device connected	USB device inserted	Low (investigate if unexpected)

Systemd Journal — Modern Linux Logging

On systemd-based systems (Ubuntu 16.04+, CentOS 7+, RHEL 7+), use journalctl:

Command	What It Does
journalctl	View all logs
journalctl -xe	View logs with explanations
journalctl -f	Follow live logs (like tail -f)
journalctl -u ssh.service	Logs for specific service
journalctl --since "2026-04-01"	Logs since date
journalctl --since "1 hour ago"	Logs from last hour
journalctl -p err	Only error-level logs
journalctl -k	Kernel logs only

journalctl -n 50	Last 50 lines

3. Essential Linux Log Analysis Commands

Basic Viewing Commands

Command	What It Does	Example
tail -f /var/log/auth.log	Follow live logs	Monitor SSH attacks in real time
head -20 /var/log/syslog	View first 20 lines	Check recent boot issues
cat /var/log/auth.log | grep sshd	View SSH-related entries	Filter by service
less /var/log/auth.log	Scroll through logs	Page through large files
zcat /var/log/auth.log.1.gz	View rotated compressed logs	Check older logs

Finding Failed SSH Attempts

Command	What It Does
grep "Failed password" /var/log/auth.log	All failed SSH attempts
grep "Failed password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c | sort -nr	Count failures per IP (sorted)
grep "Invalid user" /var/log/auth.log	Invalid username attempts
grep "Failed password for root" /var/log/auth.log | wc -l	Count root brute force attempts

Finding Successful Logins

Command	What It Does
grep "Accepted password" /var/log/auth.log	All successful SSH logins
grep "Accepted password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c	Successful logins per IP
last -a	All successful logins (from wtmp)
lastb -a	All failed logins (from btmp)
last reboot	System reboot history

Finding Sudo and Privilege Escalation

Command	What It Does
grep "sudo" /var/log/auth.log	All sudo commands
grep "COMMAND" /var/log/auth.log	Detailed sudo commands
grep "session opened for user root" /var/log/auth.log	Root access events

Finding User Management Events

Command	What It Does
grep "useradd" /var/log/auth.log	New user creation
grep "userdel" /var/log/auth.log	User deletion
grep "password changed" /var/log/auth.log	Password changes
grep "groupadd" /var/log/auth.log	New group creation

Finding Service and Process Events

Command	What It Does
grep "sshd" /var/log/auth.log	SSH service events
grep "CRON" /var/log/syslog	Scheduled task execution
grep "Failed to start" /var/log/syslog	Service failures
systemctl status sshd	Current service status

4. Section 4: Time-Based Analysis

Understanding Timestamps

Linux logs use standard date formats:

Format	Example
MMM DD HH:MM:SS	Apr 9 14:30:01
YYYY-MM-DD HH:MM:SS	2026-04-09 14:30:01
EPOCH (seconds since 1970)	1744216201

Time-Based Search Commands

Command	What It Does
grep "Apr 9 14:" /var/log/auth.log	Find logs from specific hour
sed -n '/Apr 9 14:00/,/Apr 9 15:00/p' /var/log/auth.log	Extract time range
journalctl --since "2026-04-09 14:00:00" --until "2026-04-09 15:00:00"	Time range in journalctl

5. Log Rotation and Compression

Understanding Log Rotation

Logs are rotated to prevent filling disk space. Old logs are compressed with .gz extension.

File	Content
/var/log/auth.log	Current logs (today)
/var/log/auth.log.1	Yesterday's logs
/var/log/auth.log.2.gz	Compressed log from 2 days ago

/var/log/auth.log.3.gz	Compressed log from 3 days ago

Working with Compressed Logs

Command	What It Does
zcat /var/log/auth.log.2.gz	View compressed log
zgrep "Failed password" /var/log/auth.log.2.gz	Search compressed log
zless /var/log/auth.log.2.gz	Page through compressed log

6. Detection Scenarios

Scenario 1: SSH Brute Force Attack

What to look for:

- Multiple Failed password from same IP
- Invalid user attempts
- Rapid succession of attempts

Detection command:

```
1. grep "Failed password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c | sort -nr | head -10
```

Suspicious threshold: More than 10 failures from same IP in 5 minutes.

Scenario 2: Successful Brute Force (Compromise)

What to look for:

- Multiple failures followed by Accepted password from same IP
- Login at unusual hour

Detection command:

```
1. grep "Accepted password" /var/log/auth.log | tail -20
```

Scenario 3: Privilege Escalation

What to look for:

- sudo commands from non-admin users
- session opened for user root from unexpected IP

Detection command:

```
1. grep "sudo" /var/log/auth.log | grep -v "sudo: user : TTY=pts"
```

Scenario 4: Persistence via Cron

What to look for:

- New cron jobs from /tmp or /dev/shm
- Cron jobs downloading from internet

Detection command:

```
1. grep "CRON" /var/log/syslog | grep -E "(/tmp|/dev/shm|curl|wget)"
```

Scenario 5: Malware Execution

What to look for:

- Processes running from /tmp or /dev/shm
- Suspicious command patterns

Detection command:

```
1. grep -E "(/tmp/|/dev/shm/|curl |wget |nc |netcat|python -c)" /var/log/syslog
```

Scenario 6: Data Exfiltration

What to look for:

- Large outbound connections
- Unusual data transfer patterns

Detection command:

```
1. grep -E "(scp|rsync|ftp|curl.*-T|nc.*-e)" /var/log/syslog
```

7. Forensic Artifacts

Artifact	Path	What It Contains
Bash history	~/.bash_history	User commands (per user)
Authorized keys	~/.ssh/authorized_keys	SSH public keys (persistence)
SSH config	~/.ssh/config	SSH client configuration
Known hosts	~/.ssh/known_hosts	Previously connected hosts
Cron jobs	/etc/crontab, /var/spool/cron/	Scheduled tasks (persistence)
Startup scripts	/etc/init.d/, /etc/systemd/system/	Services that start at boot
Web shell	/var/www/html/	Suspicious web files
Temp files	/tmp/, /dev/shm/	Temporary malware storage
Deleted files	(Recover with lsof | grep deleted)	Hidden malware

8. Log Analysis Commands Cheat Sheet

Task	Command
Find top attacking IPs	grep "Failed password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c | sort -nr | head -20
Find root login attempts	grep "Failed password for root" /var/log/auth.log
Find invalid usernames	grep "Invalid user" /var/log/auth.log | awk '{print $10}' | sort | uniq -c | sort -nr
Find successful logins by IP	grep "Accepted password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c | sort -nr
Find sudo commands	grep "COMMAND" /var/log/auth.log
Find user creation	grep "useradd" /var/log/auth.log
Find service failures	grep "Failed to start" /var/log/syslog
Find cron jobs from temp	grep "CRON" /var/log/syslog | grep "/tmp"
Monitor auth log live	tail -f /var/log/auth.log
View last 50 failures	tail -50 /var/log/auth.log | grep "Failed"
Search old compressed logs	zgrep "Failed password" /var/log/auth.log.2.gz
Count root login attempts	grep "Failed password for root" /var/log/auth.log | wc -l

9. Detection Rules for Linux Logs

Elastic KQL - SSH Brute Force Detection kql file:

```
event.dataset: "linux.auth" and event.type: "failure" and process.name: "sshd" and message: "*Failed password*" | count > 10 by source.ip
```

Elastic KQL - Invalid User Attempts kql file:

```
event.dataset: "linux.auth" and message: "*Invalid user*"
```

Elastic KQL - Root Login (Critical) kql file:

```
event.dataset: "linux.auth" and message: "*session opened for user root*"
```

Sigma Rule - SSH Brute Force yaml file:

```
title: SSH Brute Force Detection
logsource:
  product: linux
  service: auth
detection:
  selection:
    event.type: "failed"
    process.name: "sshd"
  threshold: 10
  time_window: 5m
  group_by: source.ip
  condition: selection
level: high
```

10. Interview Questions & Answers

Q: Where are Linux authentication logs stored?

Answer: /var/log/auth.log on Debian/Ubuntu systems, or /var/log/secure on RHEL/CentOS systems. Also journalctl for systemd-based systems.

Q: How would you detect an SSH brute force attack?

Answer: I would run grep "Failed password" /var/log/auth.log | awk '{print $11}' | sort | uniq -c | sort -nr to count failures per IP. Any IP with more than 10 failures in 5 minutes is likely a brute force attempt.

Q: What is the difference between /var/log/auth.log and /var/log/syslog?

Answer: auth.log contains only authentication-related events (logins, sudo, user management). syslog contains general system messages from all services.

Q: How do you view logs from a compressed log file?

Answer: Use zcat to view, zgrep to search, or zless to page through compressed .gz files.

Q: What commands would you use to monitor logs in real time?

Answer: tail -f /var/log/auth.log for specific files, or journalctl -f for systemd journal.

Q: How do you find when a system was last rebooted?

Answer: last reboot or uptime or who -b.

Q: What is the difference between last and lastb?

Answer: last shows successful logins (reads /var/log/wtmp). lastb shows failed logins (reads /var/log/btmp).

Q: How would you investigate a suspicious cron job?

Answer: Check /etc/crontab, /var/spool/cron/, and user-specific crontabs. Look for commands running from /tmp or /dev/shm, or downloading from internet.